

Chapter 9

Training and Adapting Vision-Language Models

Exam Study Guide, Worked Concepts, Summary-Matched Sections, Sample Questions, and Flashcards

Prepared from uploaded course materials

Main source: Chapter_9_Training_and_Adapting_VLMs.pdf
Exam source: Chapter_9_Summary_and_Sample_Questions.pdf
Practice source: Chapter_9_Flashcards.pdf

How to use it

This guide follows the same study flow used for Chapter 2: start with the quick exam map, then read the concept sections in order, then solve the sample questions, then test yourself with the flashcards. The section order is matched to the Chapter 9 summary document and expanded using the Training and Adapting VLMs lecture.

Chapter 9 Study Guide

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Most useful mental shortcut

For every Chapter 9 question, first ask: does the method change only the context, add external evidence, change trainable parameters, compare preferences, or optimize reward? Once you identify the intervention point, the equations and limitations become easier to remember.

1. Quick exam map

Chapter 9 is about adapting a pretrained VLM without losing track of what is being modified. The same autoregressive policy stays at the center; the methods differ by whether they change the prompt, evidence, parameters, preference signal, or reward objective.

Exam topic	What you must be able to say
Common policy	A VLM predicts the answer token by token: $p_{\theta}(y \mid x, v) = \prod p_{\theta}(y_t \mid y_{<t}, x, v)$.
Zero-shot	No weights are changed; only the instruction/prompt is provided.
Few-shot	No weights are changed; demonstrations alter the context and hidden-state trajectory.
RAG	Retrieved documents z are added as external evidence; the answer is a mixture over possible evidence.
SFT	Weights are updated by maximizing the likelihood of desired assistant outputs.
LoRA / QLoRA	The objective can still be SFT, but the update is restricted to low-rank adapters; QLoRA quantizes the frozen base.
DPO	Uses chosen vs. rejected answers to align the policy relative to a frozen reference policy.
RLHF / PPO / RLOO	Treats generation as a policy over tokens and optimizes a reward signal with stability controls.
Budget choice	Cheap methods are fast but temporary; weight-changing methods are reusable but more costly and riskier.

2. Summary-to-lecture matching map

This is the matching section requested: the summary document gives the exam-level storyline, while the Training and Adapting lecture gives the method-by-method details and equations. Use this table to move from the summary paragraph to the exact lecture idea you should review.

Summary document focus	Matching Training and Adapting section	Exam takeaway
Common autoregressive policy	Mathematical setup for the common VLM policy	Keep $p_{\theta}(y \mid x, v)$ fixed as the common frame for every method.
Zero-shot and few-shot keep θ fixed	Method - Zero-shot prompting; Method - Few-shot / in-context learning	They modify context, not weights.
RAG as external memory	Retrieval-Augmented Generation / field-guide retrieval section	RAG changes what evidence the model sees.
SFT as next-token maximum likelihood	Supervised fine-tuning section	SFT changes parameters by increasing likelihood of desired assistant answers.
LoRA / QLoRA as parameterization	LoRA and QLoRA sections	LoRA is not a separate task objective; it restricts the update form.
DPO as preference tuning	DPO preference dataset and DPO trainer sections	DPO compares preferred and rejected answers against a reference model.
RL-based adaptation	Reward modeling, PPO intuition, and RLOO section	RL treats answer generation as trajectories scored by rewards.
Budget ladder	Method table, empirical comparison, and final method-selection discussion	Select the least costly method that addresses the failure mode.

3. The common autoregressive VLM policy

A modern autoregressive VLM reads a text instruction x , a visual input v , and then generates an answer y one token at a time. This same policy is used before and after adaptation; what changes is either the context or the trainable parameters.

Core policy

$p_{\theta}(y \mid x, v) = \prod_{t=1}^T p_{\theta}(y_t \mid y_{<t}, x, v)$	
Symbol	Meaning
x	text prompt, instruction, or textual context
v	visual tokens/features after image preprocessing and encoding
y	full generated answer sequence
y_t	token generated at step t
$y_{<t}$	all previously generated tokens before step t
T	answer length in tokens
θ	trainable VLM parameters

Exam distinction

Do not confuse changing the text or image context with changing the model parameters. In zero-shot, few-shot, and most RAG setups, θ is fixed. In SFT, LoRA/QLoRA, DPO, and RL-based alignment, some trainable parameters or adapters are updated.

4. The adaptation ladder: what exactly changes?

The lecture organizes adaptation as a ladder. The ladder is not only weak-to-strong; it is a map of intervention types. Methods can be combined because they operate at different layers.

Layer	Method	What changes?	Persistent?
Prompt layer	Zero-shot	Instruction wording and output format only	No
Context layer	Few-shot	Demonstration examples inside the prompt	No
Evidence layer	RAG	Retrieved documents added to context	Usually no
Policy layer	SFT	Model likelihood of labeled outputs	Yes
Parameterization layer	LoRA / QLoRA	Only low-rank adapter weights move	Yes
Preference layer	DPO	Relative preference between chosen/rejected outputs	Yes
Reward layer	RLHF / PPO / RLOO	Expected reward over sampled completions	Yes

5. Zero-shot prompting

Zero-shot prompting is the cheapest baseline. The model receives an image and instruction, but no examples, no retrieved documents, and no weight update. It works only if the pretrained VLM already has

enough relevant visual and linguistic knowledge.

$$\hat{y} = \arg \max_y p_{\theta}(y \mid x, v), \quad \theta \text{ fixed}$$

- Best use: fast baseline and sanity check.
- Strength: almost free and requires no labeled dataset.
- Weakness: highly prompt-sensitive and cannot learn new domain-specific distinctions.
- Rare-species example: if the model has not learned subtle species differences, prompting alone cannot create that knowledge.

Memory line: zero-shot changes the question, not the model.

6. Few-shot / in-context learning

Few-shot prompting keeps θ fixed but adds demonstration examples. The examples teach the output format, disambiguate the task, restrict the label space, and provide a temporary task pattern inside the context window.

$$D_{ctx} = \{(x_k, v_k, y_k^*)\}_{k=1}^K$$

$$p_{\theta}(y \mid x, v, D_{ctx}) = \prod_{t=1}^T p_{\theta}(y_t \mid y_{<t}, x, v, D_{ctx})$$

- Format induction: the model infers the required output schema from examples.
- Task disambiguation: examples clarify whether to classify, explain, rank, or provide structured evidence.
- Label-space steering: repeated labels nudge the model toward the intended candidate set.
- Limitation: every example consumes context tokens and image-processing budget.

Important: few-shot is not gradient-based learning. When the prompt disappears, the adaptation disappears.

7. Retrieval-Augmented Generation (RAG)

RAG adds external memory. Instead of relying only on the pretrained model and prompt, a retriever selects relevant documents z and places them into the generation context. This is most useful when the failure is missing knowledge or weak grounding.

RAG mixture over retrieved evidence

$$p_{\theta, \eta}(y \mid x, v) = \sum_{z \in \mathcal{Z}} p_{\eta}(z \mid q(x, v)) p_{\theta}(y \mid x, v, z)$$

Posterior responsibility of a retrieved document

$$w(z) = \frac{p_{\eta}(z \mid q(x, v)) p_{\theta}(y^* \mid x, v, z)}{\sum_{z'} p_{\eta}(z' \mid q(x, v)) p_{\theta}(y^* \mid x, v, z')}$$

- $p_{\eta}(z \mid q(x, v))$ says which document the retriever thinks is relevant.
- $p_{\theta}(y \mid x, v, z)$ says how likely the generator is to answer using that evidence.
- $w(z)$ separates retrieval plausibility from explanatory usefulness.
- RAG is often a principled first intervention when the model lacks domain knowledge.

Exam sentence: RAG changes what the model sees; SFT changes how the model maps what it sees to outputs.

8. Supervised Fine-Tuning (SFT)

SFT updates model parameters using labeled triples. In multimodal instruction tuning, the model usually should learn to predict the assistant answer, not the system prompt, user prompt, or padding. This is why the loss mask is important.

$$L_{\text{SFT}}(\theta) = - \sum_{i=1}^N \sum_{t=1}^{T_i} m_{i,t} \log p_{\theta}(y_{i,t}^* \mid y_{i, < t}^*, x_i, v_i)$$

$m_{\{i,t\}}$	Meaning
1	The token contributes to the loss, usually an assistant target token.
0	The token is ignored, usually a system/user/padding token.

- SFT creates a reusable adapted model.
- It is direct imitation learning from labeled outputs.
- It can overfit or imitate noisy labels if the training set is weak.
- It can cause forgetting if the update is too aggressive.

9. LoRA and QLoRA

LoRA and QLoRA are parameter-efficient ways to implement fine-tuning. They do not replace the SFT objective; they restrict which parameters move and how much memory is needed.

$$W' = W + \Delta W, \quad \Delta W = \frac{\alpha}{r} BA$$

$$A \in \mathbb{R}^{r \times d_{in}}, \quad B \in \mathbb{R}^{d_{out} \times r}, \quad r \ll \min(d_{in}, d_{out})$$

- LoRA freezes the original weight W and trains only low-rank matrices A and B .
- Attention projections q_proj , k_proj , v_proj , and o_proj affect information routing.
- MLP projections $gate_proj$, up_proj , and $down_proj$ affect nonlinear transformation.
- QLoRA stores the frozen base model in 4-bit quantized form and trains adapters in higher precision such as bf16.
- Rank r controls adapter capacity: too small may underfit; too large costs more memory and can overfit.

Exam distinction: SFT is the training objective; LoRA is the parameter-update form.

10. Preference alignment: reward models and DPO

Preference tuning uses pairs of answers: a preferred answer y_w and a rejected answer y_l . The goal is not simply to imitate one gold answer, but to make the policy prefer better responses relative to a reference model.

Reward-model pairwise loss

$$L_{RM}(\phi) = -\log \sigma(r_\phi(x, v, y_w) - r_\phi(x, v, y_l))$$

DPO loss

$$L_{DPO}(\theta) = -\log \sigma(\beta[(\log \pi_\theta(y_w | x, v) - \log \pi_\theta(y_l | x, v)) - (\log \pi_{ref}(y_w | x, v) - \log \pi_{ref}(y_l | x, v))])$$

- DPO avoids online rollouts and PPO-style advantage estimation.
- π_{ref} anchors the policy so it does not drift too far from the original model.
- β controls the strength or sharpness of the preference margin.
- Preference alignment is not the same as factual correctness if the preferences reward style more than truth.

11. RL-based adaptation: PPO and RLOO

RL-based adaptation treats generation as a sequence of actions. The policy samples an answer, receives a scalar reward, and updates itself to make better completions more likely while avoiding unstable jumps away from the reference model.

RL term	VLM meaning
State s_t	prompt + image + generated prefix
Action a_t	next token
Policy $\pi_\theta(a_t s_t)$	next-token distribution
Trajectory	full generated answer
Reward	scalar score from reward model, verifier, or rule
Advantage A_t	how much better the outcome was than baseline

PPO ratio and clipped objective

$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$$

$$L_{clip}(\theta) = \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

RLOO baseline and loss

$$b_i = \frac{1}{G-1} \sum_{j \neq i} r_j, \quad A_i = r_i - b_i$$

$$L_{\text{RLOO}}(\theta) = -\frac{1}{G} \sum_{i=1}^G A_i \log \pi_{\theta}(o_i \mid q)$$

- PPO clipping prevents overly large policy updates.
- RLOO compares several completions from the same prompt against a leave-one-out baseline.
- RL can optimize behavior beyond imitation, but reward design is risky.
- Reward hacking occurs when the model exploits the reward rule instead of becoming genuinely better.

12. Method selection and budget ladder

The practical exam skill is choosing the right method for a failure mode. Start with the cheapest intervention that directly attacks the problem.

Failure mode	Best first method	Reason
Need quick baseline	Zero-shot	No data or training required.
Wrong output format	Few-shot	Demonstrations teach schema and style.
Missing domain knowledge	RAG	Adds external evidence without changing weights.
Need reusable task behavior	SFT or LoRA-SFT	Changes model behavior persistently.
Limited GPU memory	QLoRA	Quantized base + trainable adapters reduces VRAM.
Outputs are fluent but not preferred	DPO	Uses chosen/rejected pairs to align preferences.
Need optimize measurable reward	RLOO / PPO	Uses reward over sampled completions.

13. Common exam traps

- Saying few-shot “trains” the model. It does not update θ ; it changes context.
- Treating RAG as guaranteed truth. Retrieval can be irrelevant or misleading.
- Confusing SFT and LoRA. SFT is the objective; LoRA is the update parameterization.
- Forgetting the SFT mask $m_{\{i,t\}}$. User/system/padding tokens should usually not define the loss.
- Assuming DPO guarantees factuality. It optimizes preferences, which may reward style instead of truth.
- Ignoring π_{ref} in DPO/RLHF. The reference model prevents uncontrolled drift.
- Thinking bigger LoRA rank is always better. Rank is a capacity-vs-memory tradeoff.
- Forgetting reward hacking in RL. A model can maximize a bad reward while becoming less correct.

14. Exam-style sample questions with answers

Question 1

Why is the adaptation ladder better understood as different intervention types rather than a simple weak-to-strong ranking?

Sample answer

Because the methods modify different parts of the system. Zero-shot and few-shot modify the prompt/context, RAG modifies available evidence, SFT modifies likelihood over labeled outputs, LoRA modifies the parameterization of updates, DPO modifies preference supervision, and RL-based methods optimize reward. They can therefore be combined rather than treated as mutually exclusive.

Question 2

Why does the chapter emphasize adapting both attention projections and MLP projections in transformer-based VLMs?

Sample answer

Attention projections such as `q_proj`, `k_proj`, `v_proj`, and `o_proj` control information routing: what is queried, matched, attended, and returned to the residual stream. MLP projections such as `gate_proj`, `up_proj`, and `down_proj` control nonlinear transformation after information arrives. Many downstream VLM tasks need both new routing and new transformation.

Question 3

How does classical RL notation map onto autoregressive VLM generation?

Sample answer

The state is the prompt, image, and generated prefix; the action is the next token; the policy is $\pi_\theta(a_t | s_t)$; a trajectory is the full generated answer; and the reward is a scalar score assigned to that answer. This lets policy-gradient methods optimize token generation under multimodal context.

Question 4

Why might RAG be a more principled first intervention than SFT or DPO when a VLM lacks domain knowledge?

Sample answer

If the core failure is missing evidence, RAG directly injects relevant external documents at inference time. SFT teaches imitation and DPO adjusts preferences, but neither necessarily gives explicit supporting evidence during generation. RAG changes what the model sees.

Question 5

How can RAG, LoRA-based SFT, and RL alignment be viewed as three layers of one system?

Sample answer

RAG acts on the evidence layer: it changes what information is available. LoRA-based SFT acts on the policy-parameter layer: it changes how the model maps context to outputs. RL alignment acts on the behavioral objective layer: it changes what outputs are rewarded. These layers can be composed in one pipeline.

15. Flashcards for rapid review

The following cards are cleaned from the Chapter 9 flashcard deck and organized as a rapid review section. Redundant wording was removed while keeping the exam concepts intact.

Card 1 Q: In the context of VLMs, what does x represent in $p_{\theta}(y \mid x, v)$? A: x is the text prompt, instruction, or textual context provided to the model.	Card 2 Q: What does the visual input v represent in a standard autoregressive VLM policy? A: v is the visual tokens or features produced after image preprocessing and image encoding.
Card 3 Q: How is the output token sequence y factorized in an autoregressive VLM? A: $p_{\theta}(y \mid x, v) = \prod_{t=1}^{ T } p_{\theta}(y_t \mid y_{<t}, x, v)$.	Card 4 Q: What defines zero-shot adaptation regarding the model parameters θ? A: θ remains fixed; the model parameters are not updated during adaptation.
Card 5 Q: How does few-shot prompting alter behavior without changing weights? A: It changes the conditioning information by adding a demonstration set D_{ctx} to the prompt.	Card 6 Q: In few-shot learning, what is format induction? A: The model uses provided examples to infer the required structured output schema for the new query.
Card 7 Q: What is the main context-window constraint of few-shot prompting? A: Each extra example consumes text tokens and image-processing budget, increasing latency and memory cost.	Card 8 Q: What is Retrieval-Augmented Generation (RAG)? A: RAG augments a model with external memory by retrieving relevant documents and including them in the generation context.
Card 9 Q: In RAG, what does $p_{\eta}(z \mid q(x,v))$ represent? A: The probability assigned by the retriever to document z given the query representation $q(x,v)$.	Card 10 Q: How does RAG calculate the final answer probability as a mixture over documents? A: $p_{\{\theta,\eta\}}(y \mid x,v) = \sum_{z \in Z} p_{\eta}(z \mid q(x,v)) p_{\theta}(y \mid x,v,z)$.
Card 11 Q: In differentiable RAG, what is the posterior responsibility weight $w(z)$ used for? A: It measures how much responsibility document z receives for explaining the gold answer after evidence is considered.	Card 12 Q: What is the primary objective of supervised fine-tuning (SFT)? A: To update θ so that the desired target output sequence y^* becomes more likely for the task distribution.

Card 13 Q: Write the standard masked SFT language-model loss. A: $L_{\text{SFT}}(\theta) = -\sum_i \sum_t m_{\{i,t\}} \log p_{\theta}(y^*_{\{i,t\}} y^*_{\{i,<t\}}, x_i, v_i)$.	Card 14 Q: Why is a loss mask $m_{\{i,t\}}$ used during SFT for instruction-following VLMs? A: To ensure the model learns to predict assistant tokens while ignoring system/user prompt tokens and padding.
Card 15 Q: What is the parameter-score term in SFT updates? A: It is the gradient of the log-probability of the target sequence with respect to parameters: $\nabla_{\theta} \log p_{\theta}(y x, v)$.	Card 16 Q: How does LoRA modify a weight matrix W during fine-tuning? A: It adds a low-rank correction: $W' = W + \Delta W$, where $\Delta W = (\alpha/r)BA$.
Card 17 Q: In LoRA, what are the dimensions of A and B for $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$? A: $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$, with r much smaller than d_{in} and d_{out}.	Card 18 Q: Why are attention projections common targets for LoRA? A: They control token interaction and information routing, which are critical for new visual-semantic associations.
Card 19 Q: What is the fundamental difference between LoRA and QLoRA? A: QLoRA stores the frozen base model in 4-bit quantized form while keeping adapters in higher precision, often bf16.	Card 20 Q: In RL for VLMs, what is the state s_t during generation? A: s_t is the text prompt, image input, and all tokens generated before time step t.
Card 21 Q: What is return G_t in episodic RL for text generation? A: G_t is the total discounted sum of rewards over the generated trajectory.	Card 22 Q: What is the core idea of Direct Preference Optimization (DPO)? A: DPO optimizes the policy to rank a preferred answer y_w higher than a rejected answer y_l without an explicit reward model.
Card 23 Q: How is DPO implicit reward conceptually defined? A: $r(x, y) = \log \pi_{\theta}(y x) - \log \pi_{\text{ref}}(y x)$.	Card 24 Q: What is the role of the reference model π_{ref} in DPO? A: It anchors training and prevents the policy π_{θ} from drifting too far from the original distribution.
Card 25 Q: What is the sigmoid-margin intuition in DPO? A: DPO behaves like logistic ranking; it rewards making the chosen answer more likely than the rejected answer relative to the reference model.	Card 26 Q: In RLHF, what is the role of the reward model r_{ϕ}? A: It acts as a scalar judge that predicts a human-preference score for a prompt-answer pair.

Card 27 Q: Write the standard pairwise reward-model loss. A: $L_{RM}(\phi) = -\log \sigma(r\phi(x,v,y_w) - r\phi(x,v,y_l))$.	Card 28 Q: Why is a KL penalty added during RLHF policy optimization? A: To prevent reward hacking by discouraging the current policy from moving too far from a reference policy.
Card 29 Q: How is the RLOO advantage A_i calculated? A: $A_i = r_i - (1/(G-1))\sum_{j \neq i} r_j$, comparing one completion reward to the average reward of the other completions for the same prompt.	Card 30 Q: What does a positive advantage $A_i > 0$ imply in RLOO? A: The update should increase the log-probability of that completion.
Card 31 Q: How does the budget ladder rank SFT, LoRA, and RLHF by engineering cost? A: SFT and LoRA are medium-cost methods; RLHF has the highest engineering cost and complexity.	Card 32 Q: Which method is the fastest baseline with minimal cost and no exemplars? A: Zero-shot prompting.
Card 33 Q: What is the main advantage of RAG for grounded explanations? A: It injects external knowledge and can ground the answer in retrieved evidence.	Card 34 Q: Which VLM adaptation method is single-GPU friendly and often gives gains similar to full SFT? A: LoRA / QLoRA.
Card 35 Q: In Qwen2.5-VL-style transformers, what do MLP projections mainly control? A: They control local nonlinear feature transformation after information has been routed by attention.	Card 36 Q: Why is catastrophic forgetting a risk during aggressive SFT? A: The model may lose broader pretrained capabilities by over-optimizing for a narrow task distribution.
Card 37 Q: In DPO, what does a large β control? A: It controls the strength/sharpness of preference optimization and how strongly deviation from the reference is penalized.	Card 38 Q: What is online reinforcement learning in RLHF? A: The policy samples its own rollouts during training, so the reward signal is applied to current-policy behavior.
Card 39 Q: How does the leave-one-out baseline help in RLOO? A: It gives a lower-variance relative baseline by comparing several completions from the same prompt.	Card 40 Q: What is the significance of LoRA rank r as a modeling choice? A: Too small a rank lacks capacity; too large a rank increases memory use and may reduce regularization benefits.

Card 41 Q: According to the budget ladder, when is RAG recommended? A: When the goal is to inject specific external knowledge and provide grounded explanations.	Card 42 Q: What is implicit Bayesian-style adaptation in few-shot learning? A: The model uses demonstrations to infer a latent task description during the forward pass, adapting without gradient updates.
Card 43 Q: How does PPO clipping maintain stability? A: It clips the policy probability ratio so updates cannot change the model too aggressively in one step.	Card 44 Q: Why is dataset design as important as optimizer design in VLM tuning? A: The same data can be formatted as SFT targets, rationales, or preference pairs, each creating a different supervision signal.
Card 45 Q: What is the difference between standard RAG and RAG with backpropagation? A: Standard RAG usually uses frozen encoders; differentiable RAG updates retriever and generator jointly to improve answer likelihood.	Card 46 Q: Why is QLoRA often preferred on consumer-grade GPUs? A: 4-bit quantization reduces VRAM requirements, making large-model tuning feasible on single GPUs.
Card 47 Q: What happens mathematically if $m_{\{i,t\}}=0$ for a token? A: That token contributes nothing to the loss and produces no gradient update.	Card 48 Q: In L_{RLOO}, what does $\log \pi_{\theta}(o_i q)$ represent? A: The log-probability the current model assigns to the i-th generated completion o_i for query q.
Card 49 Q: What is the role of special control markers in a chat template? A: They mark boundaries between system, user, and assistant roles so multimodal conversations are formatted correctly.	Card 50 Q: What extra labeled data does full SFT require compared with zero-shot? A: Full SFT typically needs thousands of labeled examples, often around 5k-20k, while zero-shot needs none.
Card 51 Q: What is the implicit reward margin M in DPO used to determine? A: It measures how strongly the policy prefers the chosen answer over the rejected answer compared with the reference model.	Card 52 Q: How does a differentiable retriever in end-to-end RAG improve performance? A: It lets generation loss backpropagate into the retriever, teaching it which documents actually help the answer.
Card 53 Q: Give an example of reward hacking in the rare-species case study. A: The model may produce a nicely formatted but factually wrong answer that satisfies a superficial reward rule.	Card 54 Q: What is the main zero-shot limitation for distinguishing very similar species? A: The model cannot acquire new fine-grained distinctions beyond what it learned during pretraining.

Card 55 Q: What does the policy-gradient view of RLOO focus on? A: Increasing probabilities of high-reward completions and decreasing probabilities of low-reward completions relative to a baseline.	Card 56 Q: Why is padding ignored in SFT cross-entropy? A: Padding is not actual task data and should not affect the learned output distribution.
Card 57 Q: How does low-rank adaptation stabilize training on limited hardware? A: It greatly reduces trainable parameters, memory overhead, and overfitting risk in small-data settings.	Card 58 Q: What does advantage A_i mean conceptually? A: It measures how much better or worse an action/trajectory was than the baseline expectation.
Card 59 Q: Why does RAG aggregate documents as a weighted sum $\sum_i w_i \text{doc}_i$? A: It lets the generator condition on multiple retrieved passages while prioritizing more relevant evidence.	Card 60 Q: In QLoRA, what precision are LoRA adapters A and B typically trained in? A: They are usually trained in higher precision such as bf16 for numerical stability.
Card 61 Q: How does few-shot prompting resolve task ambiguity? A: Demonstrations clarify expected style, scientific granularity, and the required level of evidence.	Card 62 Q: In DPO, what happens if the policy already prefers the chosen answer more than the reference model does? A: The margin is large and positive, so the DPO loss becomes very small.
Card 63 Q: What is task definition in the zero-shot inference pipeline? A: It is deciding exactly what the user wants the model to do before constructing the prompt.	Card 64 Q: Which hardware was noted as capable of running LoRA training in 2-8 GPU-hours for the 3B VLM? A: An NVIDIA A100 80GB GPU.
Card 65 Q: In RLHF, what does the KL-regularized reward term do? A: It subtracts a penalty based on divergence between the current policy and the reference policy.	Card 66 Q: What is a terminal step in an episodic RL rollout? A: It is the final step in a generated sequence, usually indicated by an EOS token.
Card 67 Q: In the VLM hidden-state view, what does h_t represent? A: h_t is the hidden representation at time t used to predict the next token, influenced by both weights and context.	Card 68 Q: Why is RLOO described as simpler than PPO for VLM training? A: It avoids value networks and PPO clipping machinery while still using a stable relative baseline.

Card 69

Q: What is the main role of the retriever in the RAG inference loop?

A: It searches a corpus for the top-K documents most similar to the user query representation.

Card 70

Q: What happens if SFT uses noisy or wrong labels?

A: The model learns to imitate that noise, causing incorrect or inconsistent behavior after adaptation.

16. Last-minute checklist

- Can I write $p_{\theta}(y \mid x, v)$ and explain x , v , y , $y_{<t}$, $y_{\leq t}$, T , and θ ?
- Can I sort methods into no-weight-update and weight-update groups?
- Can I explain why few-shot changes behavior even though θ is fixed?
- Can I write the RAG mixture and posterior responsibility equations?
- Can I explain why the SFT mask $m_{\{i,t\}}$ matters?
- Can I explain LoRA dimensions and why QLoRA saves VRAM?
- Can I explain DPO using chosen/rejected answers and π_{ref} ?
- Can I map VLM generation into RL terms: state, action, policy, trajectory, reward, advantage?
- Can I explain why PPO clips and why RLOO uses a leave-one-out baseline?
- Can I choose the best first method for a failure mode: missing knowledge, wrong format, preference issue, or reward optimization?